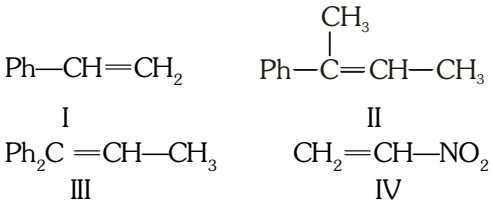
EXERCISE-I (Conceptual Questions) — Build Up Your Understanding

ELECTROPHILIC ADDITION REACTION

1. Correct reactivity order for EAR of following compounds is



(1) IV > I > II > III (2) III > II > I > IV
(3) II > III > I > IV (4) II > III > IV > I

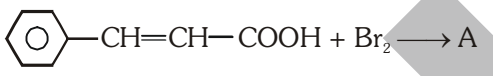
HC0160

2. The intermediate during the addition of HCl to propene in the presence of peroxide is

(1) $CH_3\dot{C}HCH_2Cl$ (2) $CH_3\overset{\oplus}{C}H—CH_3$

(3) $CH_3—CH_2—\overset{\oplus}{C}H_2$ (4) $CH_3\dot{C}HCH_3$

HC0161

3. 

the number of chiral carbons in 'A' are
(1) 1 (2) 2
(3) 3 (4) 4

HC0163

4. $CH_3—CH=CH_2 \xrightarrow{dil/H_2SO_4} A$

$CH_3—CH=CH_2 \xrightarrow{B_2H_6} \xrightarrow[OH^-]{H_2O_2} B$

Wrong statement about the product is
(1) A and B have the same functional group
(2) A and B are position isomers.
(3) A and B show chain isomerism
(4) Mixed ether is the isomer of both A and B

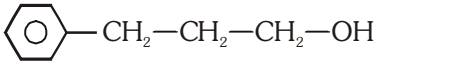
HC0164

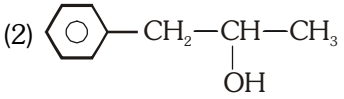
5. Which of the following alkene is most reactive for hydration
(1) ethene (2) propene
(3) 1–butene (4) 2–methyl propene

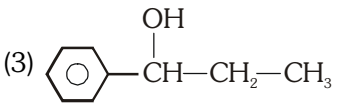
HC0165

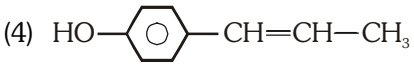
6. The major product of the following reaction is



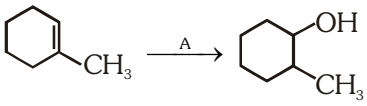
(1) 

(2) 

(3) 

(4) 

HC0166

7.  ; Reagent 'A' is

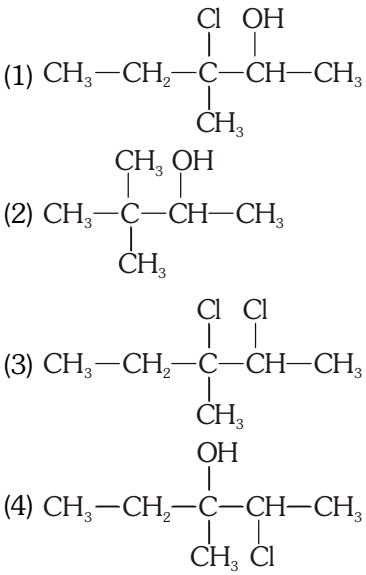
(1) $BH_3/H_2O_2/^{\ominus}OH$
(2) $H_2O/H^{\oplus}$
(3) $Hg(OCOCH_3)_2/NaBH_4$
(4) Cl_2/aq. NaOH

HC0167

8. Which of the following alkenes on hydration gives a tertiary alcohol
(1) 2–Butene (2) Isobutylene
(3) Ethene (4) α–Butylene

HC0168

9. The predominant product formed when 3–methyl–2–pentene reacts with HOCl is



HC0169

10. Propene on addition with HI, gives
(1) $CH_3—CHI—CH_3$ (2) $CH_3—CH_2—CH_2I$
(3) $CH_3—CHI—CH_2I$ (4) None of the above

HC0170

ALLEN Session : 2020-21

11. What is the main product of this reaction?

$CH_3 - C \equiv CH \xrightarrow[\text{Excess}]{HCl(g)} ?$

(1) $CH_3 - \overset{\overset{Cl}{|}}{C} = CH_2$

(2) $CH_3 - \underset{Cl}{\overset{|}{CH}} - \underset{Cl}{\overset{|}{CH_2}}$

(3) $CH_3 - CH_2 - CH\begin{matrix}Cl\\Cl\end{matrix}$

(4) $CH_3 - \underset{Cl}{\overset{\overset{Cl}{|}}{C}} - CH_3$

HC0171

12. 3–Phenyl propene on reaction with HBr gives (as a major product)

(1) $C_6H_5CH_2CH(Br)CH_3$

(2) $C_6H_5CH(Br)CH_2CH_3$

(3) $C_6H_5CH_2CH_2CH_2Br$

(4) $C_6H_5CH(Br)CH{=}CH_2$

HC0173

NUCLEOPHILIC ADDITION REACTION

13. Cyanohydrin of the following compound on hydrolysis gives compound that can show optical isomerism :

(1) HCHO (2) CH_3CHO

(3) CH_3COCH_3 (4) All the above

CC0175

14. When acetone reacts with Grignard reagent followed by hydrolysis, it gives :

(1) 1° –alcohol (2) 2° –alcohol

(3) 3° –alcohol (4) Methyl alcohol

CC0178

FREE RADICAL ADDITION REACTION

15. Reaction of HBr with propene in the presence of peroxide gives

(1) 3–bromo propane (2) Allyl bromide

(3) n–propyl bromide (4) Isopropyl bromide

HC0179

16. Isobutene $\xrightarrow[H_2O_2]{HBr}$ "product". The product is

(1) Isobutyl bromide (2) Tert. butyl bromide

(3) Tert. butyl alcohol (4) isobutyl alcohol

HC0180

FREE RADICAL SUBSTITUTION REACTION

17. The nitrating agent for the nitration of alkanes is:

(1) Conc. HNO_3

(2) Mixture of conc. HNO_3 and conc. H_2SO_4

(3) Acetyl nitrate

(4) HNO_3 vapours at high temperature

HC0182

18. The chain propagating step is fastest in the reaction of an alkane with

(1) Fluorine free radical (2) Chlorine free radical

(3) Iodine free radical (4) Bromine free radical

HC0183

19. In the nitration of propane, the product obtained in maximum yield is

(1) 1–nitropropane (2) 2-nitropropane

(3) Nitroethane (4) Nitromethane

HC0184

20. Only two isomeric monochloro derivatives are possible for (exclude stereo isomers)

(1) n-butane (2) 2, 4-dimethyl pentane

(3) benzene (4) 2-methyl butane

HC0185

21. What is the chief product obtained when n-butane is treated with bromine in the presence of light at 130°C ?

(1) $CH_3—CH_2—CH_2—CH_2—Br$

(2) $CH_3—CH_2—\underset{\underset{CH_3}{|}}{CH}—Br$

(3) $CH_3—\underset{\underset{CH_3}{|}}{CH}—CH_2—Br$

(4) $CH_3—\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}—CH_2—Br$

HC0186

ELECTROPHILIC SUBSTITUTION REACTION

22. The strongest deactivating effect on aromatic ring is

(1) $-CH_2Cl$ (2) $-OCH_3$

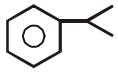
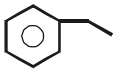
(3) $-CH_3$ (4) $-CCl_3$

AH0187

ALLEN Session 2020-21

23. Which of the following is maximum reactive towards E.S.R. :-

(1)  (2) 

(3)  (4) 

AH0188

24. Correct order of reactivity of following compound with an electrophile :-

(I) p—CH_3—C_6H_4—CH_3

(II) C_6H_5—CH_3

(III) p—CH_3—C_6H_4—NO_2

(IV) p—O_2N—C_6H_4—NO_2

(1) I > II > III > IV (2) II > I > IV > III

(3) III > II > I > IV (4) IV > III > II > I

AH0189

25. Toluene is more reactive than benzene towards electrophilic reagents due to :-

(1) Inductive effect only

(2) Hyperconjugative effect only

(3) Both inductive as well as hyperconjugative effects

(4) Strong mesomeric effect

AH0190

26. Nitration of benzene is

(1) nucleophilic substitution

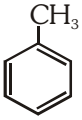
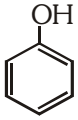
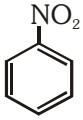
(2) nucleophilic addition

(3) electrophilic substitution

(4) electrophillic addition

AH0191

27. Consider the following compounds :

(I)  (II)  (III)  (IV) 

Correct order of their reactivity in electrophilic substitution reactions would be :-

(1) I > II > III > IV (2) IV > III > II > I

(3) III > II > I > IV (4) III > IV > I > II

AH0192

28. The active species in the nitration of benzene is

(1) NO_2^+ (2) HNO_3 (3) NO_3^- (4) NO_2^-

AH0193

29. The function of anhydrous $AlCl_3$ in the Friedel craft's reaction

(1) To absorb water

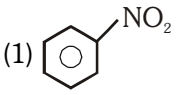
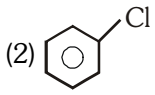
(2) To absorb HCl

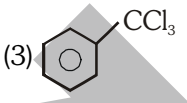
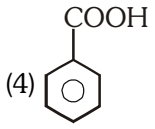
(3) To produce electrophile

(4) To produce Nucleophile

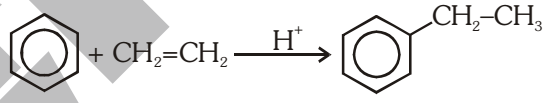
AH0194

30. In which of the following compound the electrophile attack on o– and p– positions :

(1)  (2) 

(3)  (4) 

AH0195

31. 

Incorrect statement about this reaction

(1) Benzene is substrate

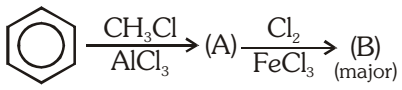
(2) Ethene is reagent

(3) Reaction is EAR with respect to ethene

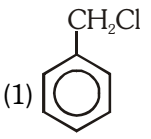
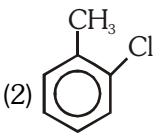
(4) Reaction is NSR for benzene

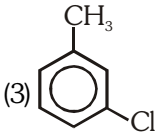
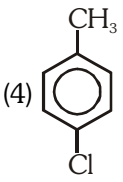
AH0401

32. For the reaction



Product B is :

(1)  (2) 

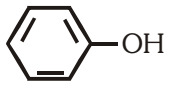
(3)  (4) 

AH0402

ALLEN Session : 2020-21

NUCLEOPHILIC SUBSTITUTION REACTION

33. The correct reactivity order of alcohols towards H–X will be

(I)  (II) $CH_2=CH-CH_2OH$

(III) CH_3-CH_2-OH (IV) $CH_3-\underset{\underset{OH}{|}}{CH}-CH_3$

(1) II > I > III > IV (2) IV > III > II > I

(3) II > IV > I > III (4) II > IV > III > I

AE0196

34. Which of the following product will be obtained when neopentyl alcohol is treated with conc. HCl in presence of $ZnCl_2$.

(1) t– butyl chloride (2) isobutylene

(3) t– pentyl chloride (4) Neo pentyl chloride

AE0197

35. In S_N1 the first step involves the formation of

(1) free radical (2) carbanion

(3) carbocation (4) final product

HD0198

36. To form alkane isonitrile, alkyl halide is reacted with:

(1) KCN (2) AgCN

(3) NaCN (4) NH_4CN

HD0199

37. Alkyl fluorides are synthesised by

(1) Finkelstein reaction (2) Swart reaction

(3) Kolbe reaction (4) Wurtz reaction

HD0201

38. The products of reaction of alcoholic silver nitrite with ethyl bromide are

(1) Ethane (2) Ethene

(3) Ethyl alcohol (4) Nitro ethane

HD0202

39. For the reaction,

$C_2H_5OH + HX \xrightarrow{ZnX_2} C_2H_5X$, the order of reactivity is

(1) HI > HCl > HBr (2) HI > HBr > HCl

(3) HCl > HBr > HI (4) HBr > HI > HCl

AE0203

40. The reaction ,$CH_3Br + OH^- \longrightarrow CH_3OH + Br^-$ obeys the mechanism

(1) S_N1 (2) S_N2

(3) E^1 (4) E^2

HD0204

41. Butanenitrile may be prepared by heating

(1) Propyl alcohol with KCN

(2) Butyl alcohol with KCN

(3) Butyl chloride with KCN

(4) Propyl chloride with KCN

HD0205

42. The given reaction is an example of

$C_2H_5Br + KCN(aq.) \longrightarrow C_2H_5CN + KBr$

(1) Elimination

(2) Nucleophilic substitution

(3) Electrophilic substitution

(4) Redox change

HD0206

43. An alkyl halide may be converted in to an alcohol by

(1) Addition (2) Substitution

(3) Dehydrohalogenation (4) Elimination

AE0208

44. Compound is most reactive towards NaOH in

(1) CH_3Cl (2) $CH_2=CHCl$

(3) C_6H_5Cl (4) $C_6H_5CH_2Cl$

HD0209

45. Most stable carbocation formed from $(CH_3)_3C-Br$, $(C_6H_5)_3CBr$,$(C_6H_5)_2CHBr$ and $C_6H_5CH_2Br$ would be

(1) $C_6H_5\overset{\oplus}{C}H_2$ (2) $(CH_3)_3\overset{\oplus}{C}$

(3) $(C_6H_5)_3\overset{\oplus}{C}$ (4) $(C_6H_5)_2\overset{\oplus}{C}H$

HD0210

46. S_N1 reaction on optically active substrates mainly gives

(1) Retention in configuration

(2) Inversion in configuration

(3) Racemised product

(4) No product

HD0211

ALLEN Session : 2020-21

47. The hydrolysis of alkyl halides by aqueous NaOH is best termed as
(1) electrophilic substitution reaction
(2) electrophilic addition reaction
(3) nucleophilic addition reaction
(4) nucleophilic substitution reaction

HD0212

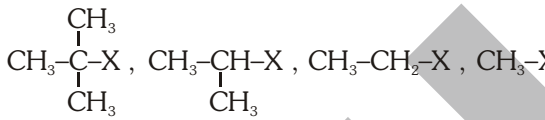
48. Reaction of sodium ethoxide and ethyl iodide will give :-
(1) Ether (2) Ethyl alcohol
(3) Acetaldehyde (4) Acetic acid

HD0213

49. The least reactive chlorine is present in
(1) Methyl chloride (2) Allyl chloride
(3) Ethyl chloride (4) Vinyl chloride

HD0214

50. Arrange the following compounds in the increasing order of their S_N^2 reactivity?



(a) (b) (c) (d)

(1) (a) < (b) < (c) < (d)
(2) (a) < (c) < (d) < (b)
(3) (d) < (c) < (b) < (a)
(4) (b) < (d) < (c) < (a)

HD0215

51. Which alcohol produces turbidity with Lucas reagent most slowly
(1) 2–Butanol (2) t–Butyl alcohol
(3) Isobutyl alcohol (4) Diphenylcarbinol

AE0216

52. The preparation of ethers from alcohols by using sulphuric acid is called :-
(1) Williamson's ether Synthesis
(2) Williamson's continuous etherification process
(3) Ziesel's method
(4) Zerewitinoff method

AE0217

53. The reaction of ethyl iodide with sodium ethoxide is
(1) An electrophilic substitution reaction
(2) A nucleophilic addition reaction
(3) A nucleophilic substitution reaction
(4) A free radical substitution reaction

HD0218

54. The Williamson synthesis involves :-
(1) A nucleophilic addition
(2) An electrophilic substitution
(3) SN^2 displacement
(4) SN^1 displacement

HD0219

55. In the Williamson synthesis of ethers given by the general equation –

$R—X + R'ONa \longrightarrow R—O—R'$ the yield from R—X follows the sequence :-
(1) $CH_3X > 1° > 2° > 3°$
(2) $CH_3X < 1° < 2° < 3°$
(3) $CH_3X < 1° < 2° > 3°$
(4) $CH_3X > 1° < 2° < 3°$

HD0220

56. With conc. HBr ethyl phenyl ether yields :–
(1) Phenol and ethyl bromide.
(2) Bromobenzene and ethanol
(3) Phenol and ethane
(4) Bromobenzene and ethane

HD0221

ELIMINATION REACTION

57. The reactivity of alkyl halides in E^2 elimination reactions follows the order
(1) R–I < R–Br < R–Cl < R–F
(2) R–F < R–Cl < R–Br < R–I
(3) R–I > R–Cl > R–Br < R–F
(4) R–I < R–Br < R–F < R–Cl

HD0223

58. The unimolecular elimination involves formation of
(1) A free radical (2) A carbanion
(3) A carbocation (4) A biradical

HD0224

ALLEN Session : 2020-21

59. Which of the following alkyl halides gives a mixture of alkenes on dehydrohalogenation
(1) n–Propyl halide
(2) Isopropyl halide
(3) s–Butyl bromide
(4) t–Butyl bromide

HD0227

60. Arrange the following alkanols A, B and C in order of their reactivity towards acid catalysed dehydration:-

(A) $CH_3-\underset{\underset{CH_3}{|}}{CH}-CH_2-CH_2-OH$

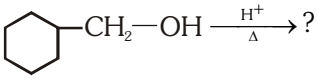
(B) $CH_3-\overset{\overset{OH}{|}}{\underset{\underset{CH_3}{|}}{C}}-CH_2-CH_3$

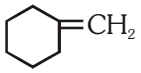
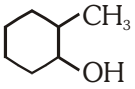
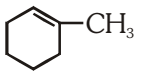
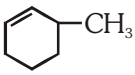
(C) $CH_3-\overset{\overset{CH_3}{|}}{CH}-\overset{\overset{OH}{|}}{CH}-CH_3$

(1) A > B > C
(2) B > A > C
(3) B > C > A
(4) C > B > A

AE0228

61. The major product in the following reaction is



(1) 
(2) 
(3) 
(4) 

HD0229

62. Arrange the following alkyl halides in decreasing order of the rate of elimination reaction with alcoholic KOH.

(A) $CH_3-\overset{\overset{H}{|}}{\underset{\underset{CH_3}{|}}{C}}-CH_2-Br$ (B) CH_3-CH_2-Br

(C) $CH_3-CH_2-CH_2-Br$
(1) A > B > C (2) C > B > A
(3) B > C > A (4) A > C > B

HD0231

ALLEN Session : 2020-21

EXERCISE-I (Conceptual Questions) ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	**2**	**2**	**2**	**3**	**4**	**3**	**1**	**2**	**4**	**1**	**4**	**2**	**2**	**3**	**3**
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	**1**	**4**	**1**	**2**	**1**	**2**	**4**	**2**	**1**	**3**	**3**	**3**	**1**	**3**	**2**
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	**4**	**2**	**4**	**3**	**3**	**2**	**2**	**4**	**2**	**2**	**4**	**2**	**2**	**4**	**3**
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	**3**	**4**	**1**	**4**	**1**	**3**	**2**	**3**	**3**	**1**	**1**	**2**	**3**	**3**	**3**
Que.	61	62													
Ans.	**3**	**4**													

EXERCISE-III (Analytical Questions) — Master Your Understanding

1. $Ph-CH=\overset{\overset{CH_3}{|}}{C}-CH_3 + HBr \xrightarrow{Peroxide} (X)$

'X' is

(1) $Ph-\underset{Br}{\overset{|}{C}}H-\underset{CH_3}{\overset{|}{C}}H-CH_3$

(2) $Ph-CH_2-\underset{CH_3}{\overset{\overset{Br}{|}}{C}}-CH_3$

(3) $Ph-CH_2-\underset{CH_3}{\overset{|}{C}}H-CH_2-Br$

(4) $Ph-CH=\underset{CH_3}{\overset{|}{C}}-CH_2-Br$

HC0335

2. $CH_3-\overset{\overset{CH_3}{|}}{C}H-CH=CH_2 \xrightarrow{\overset{\oplus}{D}/H_2O} A;$

A (major product) is

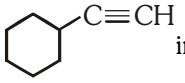
(1) $CH_3-\underset{CH_3}{\overset{|}{C}}H-CHD-CH_2-OH$

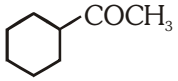
(2) $CH_3-\underset{CH_3}{\overset{|}{C}}H-\underset{OH}{\overset{|}{C}}H-CH_2-D$

(3) $CH_3-\underset{CH_3}{\overset{|}{C}}(OH)\,CHDCH_3$

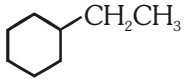
(4) $CH_3-\underset{CH_3}{\overset{|}{C}}(OH)\,CH_2-CH_2-D$

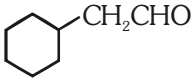
HC0337

3. Hydration of  in presence of H_2SO_4 / $HgSO_4$ gives

(1) 

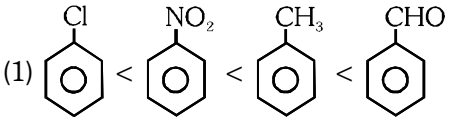
(2) 

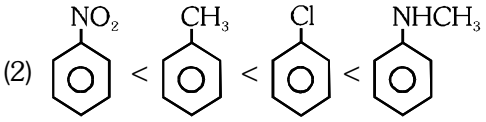
(3) 

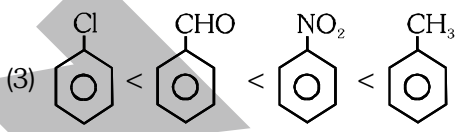
(4) 

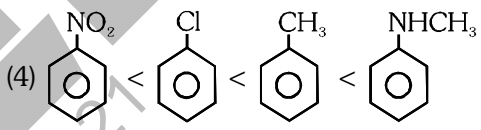
HD0339

4. Identify correct order of reactivity for electrophilic substitution reaction of the following compounds :–

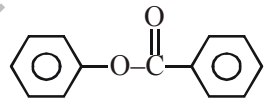
(1) 

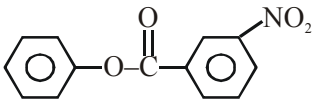
(2) 

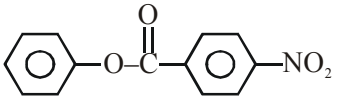
(3) 

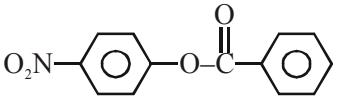
(4) 

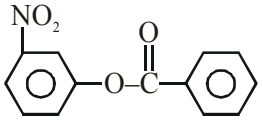
AH0340

5.  Product is :-

(1) 

(2) 

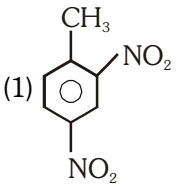
(3) 

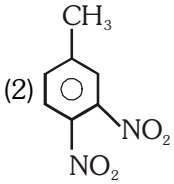
(4) 

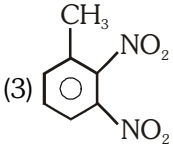
AH0341

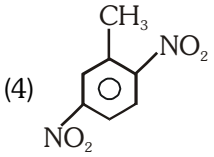
ALLEN Session : 2020-21

6. p- nitro toluene on further nitration gives

(1) 

(2) 

(3) 

(4) 

AH0342

7. Which of the following reaction gives t-butyl benzene:

(1) C_6H_6 + t–butyl chloride, $\xrightarrow{AlCl_3}$

(2) $C_6H_6 + (CH_3)_2C{=}CH_2 \xrightarrow{HF}$

(3) C_6H_6 + t–butyl alcohol $\xrightarrow{H_2SO_4}$

(4) All of them

AH0343

8. Halogenation of alkanes gives a mixture of monohalo products. The ease of substitution follows the order

(1) Tertiary H > primary H > secondary H

(2) Primary H < secondary H < tertiary H

(3) Primary H > secondary H > tertiary H

(4) Secondary H > tertiary H < primary H

GC0344

9. The number of different substitution products possible when ethane is allowed to react with bromine in sunlight are

(1) 9 (2) 6

(3) 8 (4) 5

HC0345

10. Chlorination of toluene in the presence of light and heat followed by treatment with aqueous NaOH gives

(1) o-cresol

(2) p-cresol

(3) 2, 4-dihydroxy toluene

(4) Benzyl alcohol

HC0346

11. Which gives maximum yield of C_2H_5Cl

(1) $C_2H_6 + \underset{\text{excess}}{Cl_2} \xrightarrow{hv,\ light} C_2H_5Cl + HCl$

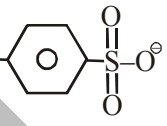
(2) $\underset{\text{excess}}{C_2H_6} + Cl_2 \xrightarrow[light]{hv} C_2H_5Cl + HCl$

(3) $C_2H_6 + Cl_2 \xrightarrow[light]{hv} C_2H_5Cl$

(4) $C_2H_6 + Cl_2 \xrightarrow{Dark} C_2H_5Cl + HCl^-$

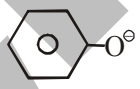
HC0347

12. Which of the following has maximum nucleophilicity :-

(1) 

(2) $CH_3–O^{\ominus}$

(3) $CF_3–\underset{\underset{O}{\|}}{C}–O^{\ominus}$

(4) 

HD0348

13. Alkaline hydrolysis of 2,2-dichloropropane gives

(1) Acetone

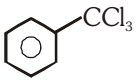
(2) 2,2-propane-diol

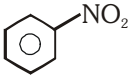
(3) Propane

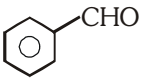
(4) Propan-2-ol

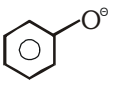
HD0350

14. Electrophile $\overset{\oplus}{N}O_2$ attacks the following :

(1) 

(II) 

(III) 

(IV) 

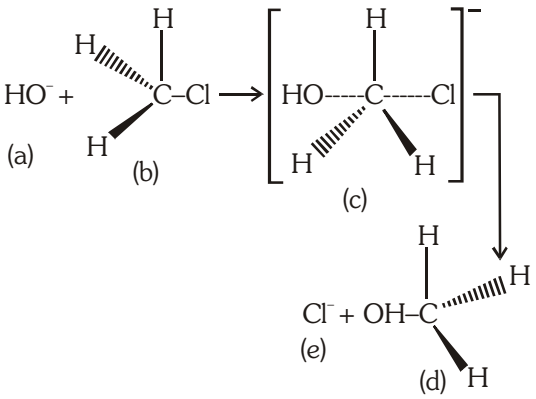
in which cases $\overset{\oplus}{N}O_2$ will attack at meta position

(1) II and IV
(2) I, II and III
(3) III and IV
(4) I only

AH0351

ALLEN Session : 2020-21

15. Which of the following statement is correct for SN^2 mechanism



(1) In (c) carbon atom is sp^3 hybridized

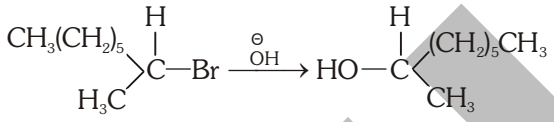
(2) In (c) carbon atom is sp^2 hybridised

(3) (a) and (e) are electrophiles

(4) (c) is more stable then (d)

HD0352

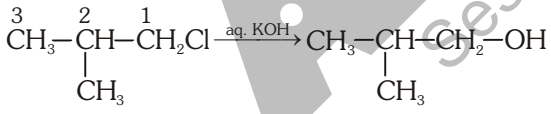
16. Following reaction is



(1) E_1 (2) S_{N^1}

(3) E_2 (4) S_{N^2}

HD0353

17. 

In the above reaction the attack of a nucleophile would be from which side

(1) On the front side of Cℓ of C_1 carbon

(2) On C_2 carbon

(3) On the rear side of Cℓ of C_1 carbon

(4) On C_3 carbon

HD0354

18. Which of the following ether can not be prepared by Williamson's method :-

(1) Ditertbutyl ether (2) Ethyl-tert-butyl ether

(3) Anisole (4) 1 & 2 Both

AE0355

19. Major product of the reaction :-

$(CH_3)_3C–Cl + C_2H_5ONa \longrightarrow$

would be :-

(1) $(CH_3)_2C–OC_2H_5$ (2) $(CH_3)_3C–C_2H_5$

(3) $(CH_3)_2C=CH_2$ (4) $CH_3–CH=CH–C_2H_5$

HD0357

20. The major product obtained on treatment of $CH_3CH_2CH(F)CH_3$ with CH_3O^-/CH_3OH is

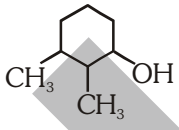
(1) $CH_3CH_2CH(OCH_3)CH_3$

(2) $CH_3CH=CHCH_3$

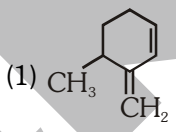
(3) $CH_3CH_2CH=CH_2$

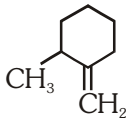
(4) $CH_3CH_2CH_2CH_2OCH_3$

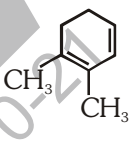
HD0359

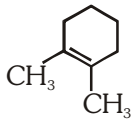
21. 

X (major) 'A' is

(1) 

(2) 

(3) 

(4) 

HD0361

22. In following two columns :

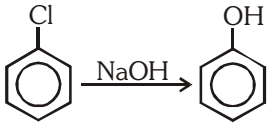
Column-I

(a) $CH_3–CH_2–Br + HS^- \rightarrow CH_3–CH_2–SH + Br^-$

(b) $(CH_3)_2C=CH_2 + HOCl \rightarrow (CH_3)_2\overset{OH}{\overset{|}{C}}–CH_2–Cl$

(c) $CH_3–CH_2–Br + {}^-OH \rightarrow CH_2=CH_2 + H_2O + Br^-$

(d) $(CH_3)_3C–CH_2–OH + H–Br \rightarrow (CH_3)_2CBrCH_2–CH_3 + H_2O$

(e) 

Column-II

(1) Nucleophilic substitution

(2) Elimination

(3) Electrophilic substitution

(4) Electrophilic additon

(5) Rearrangement reaction

(1) a, d → 1 , b → 4, c → 2 , e → 1

(2) e, a, d → 1 , b → 4, c → 3

(3) a, e → 1 , b → 4, c → 2 , d → 5

(4) a, c → 1 , b → 4, e → 3 , d → 5

HD0403

ALLEN Session : 2020-21

23. Which is not correct about nitration ?

(1) $\overset{\oplus}{N}O_2$ is an attacking electrophile

(2) In general, nitration is monosubstitution process

(3) In nitrating mix, HNO_3 acts as acid, & H_2SO_4 acts as base

(4) In nitrating mix HNO_3 acts as base & H_2SO_4 acts as an acid

AH0404

ALLEN
Session : 2020-21

EXERCISE-III (Analytical Questions) ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	**2**	**4**	**1**	**4**	**3**	**1**	**4**	**2**	**1**	**4**	**2**	**2**	**1**	**2**	**2**
Que.	16	17	18	19	20	21	22	23							
Ans.	**4**	**3**	**1**	**3**	**3**	**4**	**1**	**3**							